

MATH 162, Winter 2026
SCRIPT 9: Continuous Functions

We recall from Script 1 that if $f: X \rightarrow Y$ is a function and $B \subset Y$, then the *preimage of B under f* is the set

$$f^{-1}(B) = \{a \in X \mid f(a) \in B\}.$$

We now give some basic properties of preimages.

Lemma 9.1. *Let $X \subset \mathbb{R}$ and $f: X \rightarrow \mathbb{R}$. If $A, B \subset \mathbb{R}$, then*

$$\begin{aligned} f^{-1}(A \cup B) &= f^{-1}(A) \cup f^{-1}(B), \\ f^{-1}(A \cap B) &= f^{-1}(A) \cap f^{-1}(B), \\ f^{-1}(A \setminus B) &= f^{-1}(A) \setminus f^{-1}(B) \\ \text{and} \quad f^{-1}(\mathbb{R}) &= X. \end{aligned}$$

If $m \in f^{-1}(A \cup B)$, then $f(m) \in A \cup B$, then $f(m) \in A$ or $f(m) \in B$. Hence, $m \in f^{-1}(A)$ or $m \in f^{-1}(B)$, hence $m \in f^{-1}(A) \cup f^{-1}(B)$. This completes the first containment.

Exercise 9.2. Let $X \subset \mathbb{R}$ and $f: X \rightarrow \mathbb{R}$. Let $A \subset X$ and $B \subset \mathbb{R}$. Show that

$$f(f^{-1}(B)) \subset B \quad \text{and} \quad A \subset f^{-1}(f(A)).$$

Give examples to show that the inclusions can be proper.

Exercise 9.2. Let $X \subset \mathbb{R}$ and $f: X \rightarrow \mathbb{R}$. Let $A \subset X$ and $B \subset \mathbb{R}$. We want to show that $f(f^{-1}(B)) \subset B$ and $A \subset f^{-1}(f(A))$.

Firstly, let us consider $f(f^{-1}(B)) \subset B$. Let there be an arbitrary $y \in f(f^{-1}(B))$. As we know that $f^{-1}(B)$ may be written as $\{a \in X \mid f(a) \in B\}$, we may say that there exists some $x \in X$ where $x \in f^{-1}(B)$ such that $y = f(x)$.

Now consider $f^{-1}(B)$. By the definition of the preimage $f^{-1}(B)$, we know $f(x) \in B$. As $f(x) = y$ we may say that $y \in B$. As we took an arbitrary $y \in f(f^{-1}(B))$ and have shown that $y \in B$, we get that $f(f^{-1}(B)) \subset B$. This works to prove that $f(f^{-1}(B)) \subset B$.

Secondly, let us consider $A \subset f^{-1}(f(A))$. Let $x \in A$. We want to show that $x \in f^{-1}(f(A))$. By the definition of $f^{-1}(f(A)) = \{z \in X \mid f(z) \in f(A)\}$. Therefore, we want to show that $f(x) \in f(A)$. By the definition of a function, we can write that $f(A) = \{f(a) \mid a \in A\}$. As $A \subset X$, we may say that $x \in A$ and therefore $x \in X$. Since $x \in A$ we have that $f(x)$ is the image of an element in A (namely, x itself). Therefore, by the definition of $f(A)$, we have $f(x) \in f(A)$. Since, $f(x) \in f(A)$, and by the definition of $f(A)$, we have $f(x) \in f(A)$. As $f(x) \in f(A)$, by the definition of the pre-image, we have that $x \in f^{-1}(f(A))$. As x was arbitrary, we may say that every element of A is in $f^{-1}(f(A))$. Therefore, this works to prove that $A \subset f^{-1}(f(A))$.

□

Exercise 9.3. Let $X \subset \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$. Let $A \subset X$ and $B \subset \mathbb{R}$. Then $f(A) \subset B$ if, and only if, $A \subset f^{-1}(B)$.

First, we want to show that if $f(A) \subset B$ then $A \subset f^{-1}(B)$. We have that $m \in f(A)$ implies $m \in B$. Hence, $n \in A$ implies $f(n) \in f(A)$ implies $f(n) \in B$, hence $n \in f^{-1}(B)$. Thus, $A \subset f^{-1}(B)$.

Second, we want to show that if $A \subset f^{-1}(B)$ then $f(A) \subset B$. We have that $m \in A$ implies $m \in f^{-1}(B)$. Hence, $n \in f(A)$ implies $f^{-1}(n) \in A$ implies $f^{-1}(n) \in f^{-1}(B)$, hence $n \in B$. Thus, $f(A) \subset B$.

Definition 9.4. Let $X \subset \mathbb{R}$. A function $f : X \rightarrow \mathbb{R}$ is *continuous* if for every open set $U \subset \mathbb{R}$, the preimage $f^{-1}(U)$ is open in X .

Proposition 9.5. Let $X \subset \mathbb{R}$. A function $f : X \rightarrow \mathbb{R}$ is continuous if, and only if, for every closed set $F \subset \mathbb{R}$, the preimage $f^{-1}(F)$ is closed in X .

First, we want to show that if f is continuous, then for every closed set $F \subset \mathbb{R}$, the preimage $f^{-1}(F)$ is closed in X .

If f is continuous, then we know that for every open set $U \subset \mathbb{R}$, $f^{-1}(U)$ is open in X . Then, for every closed set $F \subset \mathbb{R}$, $f^{-1}(F) = f^{-1}(\mathbb{R} \setminus U) = f^{-1}(\mathbb{R}) \setminus f^{-1}(U)$ for some open set U (the last equality follows from Lemma 9.1).

Then, $f^{-1}(F) = f^{-1}(\mathbb{R}) \setminus f^{-1}(U) = X \setminus f^{-1}(U)$. We know that we can write $f^{-1}(U) = X \cap O$ for some open set O . Then, $f^{-1}(F) = X \setminus (X \cap O) = X \setminus O = X \cap (\mathbb{R} \setminus O)$, thus $f^{-1}(F)$ is closed in X , since O is open implies $\mathbb{R} \setminus O$ is closed.

Next, we want to show that if for every closed set $F \subset \mathbb{R}$ the preimage $f^{-1}(F)$ is closed in X , then f is continuous.

Let F be any closed set, then $\mathbb{R} \setminus F$ is open. We want to show that $f^{-1}(\mathbb{R} \setminus F)$ is open in X .

We have that $f^{-1}(\mathbb{R} \setminus F) = f^{-1}(\mathbb{R}) \setminus f^{-1}(F) = X \setminus f^{-1}(F)$. We know that $f^{-1}(F)$ can be written as $X \cap C$ for some closed set C . Then, $f^{-1}(\mathbb{R} \setminus F) = X \setminus f^{-1}(F) = X \setminus (X \cap C) = X \cap C^c = X \cap (\mathbb{R} \setminus C)$, and this completes the proof because C is closed implies $\mathbb{R} \setminus C$ is open.

Definition 9.6. Let $X \subset Y \subset \mathbb{R}$ and let $f : Y \rightarrow \mathbb{R}$. Then the *restriction of f to X* , written $f|_X$, is the function $f|_X : X \rightarrow \mathbb{R}$ defined by

$$f|_X(x) = f(x), \quad \text{for all } x \in X.$$

Proposition 9.7. Let $X \subset Y \subset \mathbb{R}$. If $f : Y \rightarrow \mathbb{R}$ is continuous, then the restriction of f to X is continuous.

It is important that the definition of continuity of $f : X \rightarrow \mathbb{R}$ states that the preimage of an open set is *open in X* but not necessarily open in $\mathbb{R}$.

Exercise 9.8. Let $X \subset \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$. Let $U \subset \mathbb{R}$ be open. Show that

- i) if X itself is open then $f^{-1}(U)$ is open in X if, and only if, it is open (in $\mathbb{R}$);
- ii) if X is not open then there is an open subset U for which $f^{-1}(U)$ is open in X but not open in $\mathbb{R}$.

Exercise 9.8, Part 1. Let $X \subset \mathbb{R}$ be an open set and $f : X \rightarrow \mathbb{R}$ be a function. Let $U \subset \mathbb{R}$ be an open set. We wish to show that $f^{-1}(U)$ is open in X if and only if it is open in $\mathbb{R}$.

($\implies$) Suppose first that $f^{-1}(U)$ is open in X . By the definition of the subspace topology, there exists a set $G \subset \mathbb{R}$ that is open in $\mathbb{R}$ such that $f^{-1}(U) = X \cap G$. To show $f^{-1}(U)$ is open in $\mathbb{R}$, let $x \in f^{-1}(U)$ be an arbitrary point. Since $x \in X \cap G$, it follows that $x \in X$ and $x \in G$. Because G is open in $\mathbb{R}$, there exists $\epsilon_G > 0$ such that the interval $(x - \epsilon_G, x + \epsilon_G) \subseteq G$. Similarly, because X is open in $\mathbb{R}$, there exists $\epsilon_X > 0$ such that the interval $(x - \epsilon_X, x + \epsilon_X) \subseteq X$. Let $\epsilon = \min(\epsilon_G, \epsilon_X)$. Since $\epsilon \leq \epsilon_G$ and $\epsilon \leq \epsilon_X$, we have $(x - \epsilon, x + \epsilon) \subseteq G$ and $(x - \epsilon, x + \epsilon) \subseteq X$. Consequently, $(x - \epsilon, x + \epsilon) \subseteq X \cap G = f^{-1}(U)$. Since for every $x \in f^{-1}(U)$ there exists an open interval around it contained in the set, $f^{-1}(U)$ is open in $\mathbb{R}$.

($\impliedby$) Conversely, suppose that $f^{-1}(U)$ is open in $\mathbb{R}$. We must show that $f^{-1}(U)$ is open in X . Recall that a set $V \subseteq X$ is open in X if there exists an open set $M \subseteq \mathbb{R}$ such that $V = X \cap M$. Since $f^{-1}(U)$ is a subset of the domain X , we can trivially write $f^{-1}(U) = X \cap f^{-1}(U)$. By our assumption, $f^{-1}(U)$ is open in $\mathbb{R}$. Thus, by setting $M = f^{-1}(U)$, we see that $f^{-1}(U)$ is the intersection of X and a set that is open in $\mathbb{R}$. By the definition of the subspace topology, $f^{-1}(U)$ is open in X . □

Proof. Suppose $X \subset \mathbb{R}$ is not open. By the definition of an open set, this implies there exists at least one point $x_0 \in X$ such that for every $\epsilon > 0$, the interval $(x_0 - \epsilon, x_0 + \epsilon)$ is not a subset of X . We wish to show there exists an open set $U \subset \mathbb{R}$ such that $f^{-1}(U)$ is open in X but not open in $\mathbb{R}$.

Consider the identity function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = x$. Let U be the open interval $(x_0 - 1, x_0 + 1)$, which is clearly open in $\mathbb{R}$. The preimage of U under f is given by $f^{-1}(U) = \{x \in X \mid x \in (x_0 - 1, x_0 + 1)\}$, which can be written as the intersection $f^{-1}(U) = X \cap (x_0 - 1, x_0 + 1)$. Since $(x_0 - 1, x_0 + 1)$ is an open set in $\mathbb{R}$, it follows from the definition of the subspace topology that $f^{-1}(U)$ is open in X .

However, we claim that $f^{-1}(U)$ is not open in $\mathbb{R}$. Suppose for the sake of contradiction that $f^{-1}(U)$ were open in $\mathbb{R}$. Since $x_0 \in f^{-1}(U)$, there would exist some $\epsilon_0 > 0$ such that the interval $(x_0 - \epsilon_0, x_0 + \epsilon_0) \subseteq f^{-1}(U)$. Because $f^{-1}(U) \subseteq X$, this would imply that $(x_0 - \epsilon_0, x_0 + \epsilon_0) \subseteq X$. This contradicts our initial assumption that x_0 is a point in X for which no such ϵ -neighborhood is contained within X . Therefore, $f^{-1}(U)$ must not be open in $\mathbb{R}$. □

Definition 9.9. Let $X \subset \mathbb{R}$. The function $f : X \rightarrow \mathbb{R}$ is *continuous at* $x \in X$ if, for every region R containing $f(x)$, there exists an open set S containing x such that $S \cap X \subset f^{-1}(R)$.

Theorem 9.10. Let $X \subset \mathbb{R}$. The function $f : X \rightarrow \mathbb{R}$ is continuous if and only if it is continuous at every $x \in X$.

Proof. Let $f : X \rightarrow \mathbb{R}$. We show that f is continuous (preimages of open sets are open in X) if and only if it is continuous at every $x \in X$.

($\implies$) Suppose f is continuous. To show it is continuous at $c \in X$, pick $\epsilon > 0$. The interval $U = (f(c) - \epsilon, f(c) + \epsilon)$ is open in $\mathbb{R}$. $f^{-1}(U)$ is open in X , meaning $f^{-1}(U) = X \cap G$ for some open $G \subset \mathbb{R}$. Since $c \in G$ and G is open, there is a $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq G$. Thus, for any $x \in X$ within δ of c , we have $x \in X \cap G$, which means $f(x) \in U$. This demonstrates that $|f(x) - f(c)| < \epsilon$.

($\impliedby$) Suppose f is continuous at every $x \in X$. Let $U \subset \mathbb{R}$ be open. To show $f^{-1}(U)$ is open in X , we show every point in $f^{-1}(U)$ has a neighborhood contained in the set. For any $c \in f^{-1}(U)$, we have $f(c) \in U$. Since U is open, there is an $\epsilon > 0$ such that $(f(c) - \epsilon, f(c) + \epsilon) \subseteq U$. there is a $\delta_c > 0$ such that f maps $X \cap (c - \delta_c, c + \delta_c)$ into U . This means $X \cap (c - \delta_c, c + \delta_c) \subseteq f^{-1}(U)$. Since $f^{-1}(U)$ is the union of these relatively open neighborhoods, it is open in X . $\square$

Next, we discuss the relationship between continuity and connectedness.

Theorem 9.11. *Let $X \subset \mathbb{R}$. Suppose that $f: X \rightarrow \mathbb{R}$ is continuous. If X is connected, then $f(X)$ is connected.*

Proof of Theorem 9.11. Let $X \subset \mathbb{R}$. Suppose $F: X \rightarrow \mathbb{R}$ is continuous. If X is connected, $f(X)$ is connected.

Assuming for the sake of contradiction, say $f(X)$ is not connected. $f(X) = A \cup B$ when A, B is open in $f(X)$, nonempty subset of $\mathbb{R}$ and $A \cap B = \emptyset$.

Because f is continuous, $f^{-1}(A)$ and $f^{-1}(B)$ are open in X . Note that $f^{-1}(A) \cup f^{-1}(B) \subset X$.

Take some $x \in X$. then $f(x) \in f(X)$. So $f(x) \in A \cup B$ so $f(x) \in A$ or $f(x) \in B$. Therefore, $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$

$f^{-1}(A)$ and $f^{-1}(B)$ are nonempty since they are open. Since A and B are nonempty, so... so there must be $\exists a', b'$ where $f(a') \in A$. and $f(b') \in B$. Therefore, $a' \in f^{-1}(A)$, and $b' \in f^{-1}(B)$

$\square$

Corollary 9.12. *Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then for every point p between $f(a)$ and $f(b)$ there exists c such that $a < c < b$ and $f(c) = p$.*

Proof of 9.12. WTS that if $f: [a, b] \rightarrow \mathbb{R}$ is continuous then $\forall p$ such that $f(a) < p < f(b) \exists c$ such that $a < c < b$ and $f(c) = p$. By 8.16 we have that $[a, b]$ is connected. By 9.11, since f is continuous and $[a, b]$ is connected we have say that $f([a, b])$ is connected as it is an interval it contains all points between $f(a)$ and $f(b)$. Therefore $\forall p$ such that $f(a) < p < f(b)$

$\square$

Finally we consider the relationship between continuity and inverse functions.

Lemma 9.13. *If $f: (a, b) \rightarrow \mathbb{R}$ is continuous and injective, then f is either strictly increasing or strictly decreasing on (a, b) .*

Proof. If $f: (a, b) \rightarrow \mathbb{R}$ is continuous and injective then f is strictly increasing or decreasing. Let $c, d \in (a, b)$ where $c < d$. Consider $[c, d]$. $f|_{[c, d]}$ is continuous by 9.7. Without loss of generality $f(c) < f(d)$. Let $p \in (f(c), f(d))$. By 9.12, $\exists x \in (c, d)$ such that $f(x) = p$. Now consider c

$\square$

In Proposition 1.27 we saw that if $f : A \rightarrow B$ is bijective then there is a bijection $g : B \rightarrow A$, called the *inverse function*, such that $(g \circ f)(a) = a$, for all $a \in A$, and $(f \circ g)(b) = b$, for all $b \in B$. We denote the inverse function g by f^{-1} .

Exercise 9.14. Show that if $f : A \rightarrow \mathbb{R}$ is injective, then there is an injection $g : f(A) \rightarrow \mathbb{R}$ such that $(g \circ f)(a) = a$, for all $a \in A$, and $(f \circ g)(b) = b$, for all $b \in f(A)$. This g is called the inverse function for f .

Theorem 9.15. If $f : (a, b) \rightarrow \mathbb{R}$ is continuous and injective, then the inverse function $g : f((a, b)) \rightarrow \mathbb{R}$ is continuous.

Additional Exercises

1. Suppose $f : X \rightarrow \mathbb{R}$. Show that f is continuous at $x \in X$ if and only if for every open set U containing $f(x)$, there is a set V that is open in X such that $x \in V$ and $f(V) \subset U$.
2. Let $X \subset \mathbb{R}$ and let $f : X \rightarrow \mathbb{R}$. Let $A, B \subset \mathbb{R}$. Either prove or give a counterexample to each of the following:
 - a) $f(A \cup B) = f(A) \cup f(B)$
 - b) $f(A \cap B) = f(A) \cap f(B)$
 - c) $f(A \setminus B) = f(A) \setminus f(B)$
 - d) $f(X) = \mathbb{R}$.
3. Let $A \subset \mathbb{R}$ be an open set and define $\chi_A : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\chi_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

- (a) Show that χ_A is continuous at every point in A .
 - (b) Where else is χ_A continuous?
4. Suppose that $f : (a, b) \rightarrow \mathbb{R}$ is a continuous function. Prove that for any $x_1, x_2, \dots, x_n \in (a, b)$ there exists $t \in (a, b)$, such that

$$f(t) = \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}.$$